

AWS Auto Scaling – Full Explanation with All Components

AWS Auto Scaling is a service that automatically adjusts the number of Amazon EC2 instances (or other resources) in response to traffic or load conditions. It helps maintain **performance, availability, and cost-efficiency** by **scaling out** (adding) or **scaling in** (removing) resources as needed.

Core Components of AWS Auto Scaling

1. Launch Template / Launch Configuration

- **Purpose:** Defines **how to launch EC2 instances**.
 - **Includes:**
 - AMI (Amazon Machine Image)
 - Instance type
 - Key pair
 - Security groups
 - IAM role
 - User data (startup script)
 - **Launch Template** is the recommended modern method (more flexible than Launch Config).
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2. Auto Scaling Group (ASG)

- **Definition:** A group of EC2 instances managed as a **single unit**.
 - **Functions:**
 - Maintains a specified number of healthy instances.
 - Automatically replaces unhealthy instances.
 - Handles scaling policies.
 - **Settings:**
 - **Minimum size:** Minimum number of instances
 - **Maximum size:** Max allowed instances
 - **Desired capacity:** Target number of instances
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3. Scaling Policies

These define **when and how** the ASG should scale in or out.

a. Dynamic Scaling

- Responds to real-time metrics (e.g., CPU > 70%).
- Uses **CloudWatch alarms** and policies.

b. Predictive Scaling

- Uses **machine learning** to forecast traffic patterns and scales in advance.

c. Scheduled Scaling

- Pre-defined scaling based on time (e.g., add 2 instances every weekday at 9 AM).
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4. Health Checks

- Ensures only **healthy instances** remain in service.
 - If an instance fails:
 - Auto Scaling automatically **terminates** it.
 - Launches a **new one** to replace it.
 - Health checks can be:
 - **EC2-based** (default)
 - **ELB-based** (if connected to a Load Balancer)
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5. Elastic Load Balancer (Optional but Common)

- Distributes traffic across the instances in the Auto Scaling Group.
 - Ensures only **healthy instances** receive traffic.
 - Commonly used with:
 - **ALB** (Application Load Balancer)
 - **NLB** (Network Load Balancer)
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6. 🔍 CloudWatch Alarms

- Monitors metrics like:
 - CPU utilization
 - Memory usage (custom metrics)
 - Request count
 - Triggers **scaling policies** when thresholds are met.
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7. 📈 Target Tracking Scaling

- Simplifies scaling with a single metric (e.g., keep CPU at 60%).
 - Auto Scaling automatically adjusts instance count to maintain the target value.
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🔄 Workflow of AWS Auto Scaling

1. Define a Launch Template
 2. Create an Auto Scaling Group with min/max/desired capacity
 3. Attach scaling policies (e.g., target tracking or dynamic rules)
 4. Set up CloudWatch alarms for scaling events
 5. Optionally attach a Load Balancer
 6. Auto Scaling launches/terminates instances as needed
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✅ Benefits of AWS Auto Scaling

Benefit	Description
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High Availability	Replaces failed instances automatically
Cost Optimization	Scales in during low demand
Better Performance	Scales out during peak load
Fully Automated	No manual intervention required
Integrated Monitoring	Works with Amazon CloudWatch

Use Cases

Use Case	Example
Web app scaling	Add EC2s when CPU > 70%
Seasonal traffic	Scale based on forecasted demand
Failover recovery	Replace failed instances in other AZs
Batch processing	Add workers during job processing

Example Configuration Summary

Setting	Value
Min Instances	2
Max Instances	10
Desired Capacity	4
Launch Template	my-app-template
Scaling Policy	Target tracking (CPU = 60%)
Load Balancer	ALB on port 80/443
